BAT54S
RD_B2
10k 0.1%
V+
OUTB
-INB
+INB

U2
OUTA
INA-
INA+
V+
INB+
INB-
OUTB
OUTD
IND-
IND+
V-
INC+
INC-
OUTC
OPA4388IDR
RF2
CF1
2.2nF C0G
R_ADC0
100R 1%
COL
TIA
SENSE2
SENSE3
GND
DRV2

-> 10k -> OPA minus; internal input-clamp current bounded, fault/thermal and macro loop HOLD

A output -> 1k -> TIA tap; all ADC/filter/feedback takeoffs after isolation

ur continuous TIAs; Rf4.99k || Cf2.2nF from isolated TIA tap to COL

2 REVIEW ONLY - Continuous_Column_TIAs - 3/6

1 SDI
2 RST#/PD#
3 DAISY
4 REFSEL#
5 REFIO
6 REFGND
7 REFCAP
8 AGND
9 AVDD
10 AUX_IN
11 AUX_GND
12 NC
13 NC
14 NC
15 NC
16 AIN_0P
17 AIN_0GND
18 AIN_1P
19 AIN_1GND
CS# 38
SCLK 37
SDO 36
DNC 35
DVDD 34
DGND 33
AGND 32
AGND 31
AVDD 30
AGND 29
AGND 28
NC 27
NC 26
NC 25
NC 24
AIN_3P 23
AIN_3GND 22
AIN_2P 21
AIN_2GND 20
ADS8684IDBTR

Effective capacitance and placement must meet datasheet: REFERENCE_CAPACITANCE_HOLD

AVDD pins9/30 each44uF+2.2uF nominal; DVDD pin34 44uF nominal

Internal4.096V ADC reference; REFIO and REFCAP independently bulk bypassed

R2 REVIEW ONLY - ADS8684_Decoupling - 4/6

U7
PB9
PC14-OSC32IN
PC15-OSC32OUT
VDD/VDDA
VSS/VSSA
PF2-NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB8
PB7
PB6
PB5
PB4
PB3
PA15
PA14-BOOT0
PA13
PA12[PA10]
PA11[PA9]
PA10
PC6
PA9
PA8
PB2
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
STM32G031K8T6
1
2
3
4
HDR-M

Offline SPI: 0-5.12V range0x06, tagged48clk frames, 40kSPS / hardware timing HOLD

External SWD/UART/sense lines all4.99k current limited, clamps and rail sink

GPIO PB1 pin16 = MCU_ENABLE; PGOOD resets MCU; Schmitt reshapes hardware gate edge

R2 REVIEW ONLY - MCU_Protected_Interfaces - 5/6

U9
EN/UVLO
OVLO
PG
PGTH
IN
LM73100RPWR
10k 0.1%
No manufacture / procu
ch power release. DYNAM
LT HOLD retain
PGOOD = both superv
utputs released; nominal
30V/3.10385V
LM73100 both supply
89001 monitors actual V
R2 REVIEW ONL
r_Qualification - 6/